Simone Rossetti

Applied Scientist / Research Engineer

Multimodal & Foundation Models · Research-to-System Translation

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Professional Summary

Multimodal and foundation model specialist with first-author publications at NeurIPS, ICCV, and ECCV. Translates research into robust systems: from methodology design (data-efficient training, cross-modal alignment, evaluation frameworks) to implementation (distributed training pipelines, model serving, CI-based validation). Combines evaluation rigor with system robustness. Industrial R&D orientation with four years leading applied research from prototype to field-validated AI systems.

Core Expertise

Multimodal & Vision-Language Systems	Foundation Model Adaptation & Fine-Tuning
Data-Efficient Training & Supervision Design	Large-Scale & Distributed Training
Evaluation & Robustness Frameworks	Research-to-Deployment Systems

Professional Experience

Co-Founder and Lead Applied Researcher at DeepPlants S.r.l. Oct 2021 – Present

- Defined technical vision for multimodal AI-driven decision-support systems and led a 5-person R&D team, advancing prototypes from TRL1 to TRL5 within 12 months.
- Designed data-efficient training strategies and implemented distributed PyTorch pipelines (Hugging Face, OpenCLIP, xFormers), reducing annotation requirements by 70% while maintaining ≥85% performance across heterogeneous conditions.
- Developed large-scale multimodal evaluation suites (∼50K VQA, ∼25K structured QA) and established reproducible training and evaluation frameworks with CI-based validation for internal R&D standards.
- Conducted masked pretraining and finetuned vision-language models (Qwen3-VL, LLaMA 3); implemented containerised serving (Ollama, vLLM, SGLang) and agentic pipelines (LangGraph, RAG) with instrumentation for latency profiling.
- Defined technical methodologies and experimental validation strategies for EU-funded research initiatives in AI-driven sustainable agriculture.

Research Fellow at AlcorLAB – Sapienza University (DIAG) Jan 2021 – Oct 2021

- Designed and implemented distributed multi-GPU training pipelines for spatiotemporal models on AVA and YouTubeVIS benchmarks.
- Conducted systematic ablation studies to quantify architectural trade-offs under memory and latency constraints, establishing evaluation protocols for cross-dataset robustness.

Education

2021 – 2025	PhD in Computer Science Engineering at **Sapienza University**	(Excellent)
2019 – 2021	MSc in Artificial Intelligence and Robotics at **Sapienza University**	(110/110 laude)
2015 – 2019	BSc in Computer Engineering at **Roma Tre University**	(95/110)

Selected Projects & Systems

Cabbo	AI-powered agricultural decision-support system (DeepPlants) for farm management; web and mobile deployment for rural accessibility.
CABBAGE	Large-scale multimodal benchmark for AI in agriculture: Visual Cognition, Scientific Knowledge, Procedural Reasoning.
ViT-PCM	Weakly supervised semantic segmentation (ECCV, ICCV); adapted for plant phenotyping in vertical farming.
HAUS	Unsupervised semantic segmentation (NeurIPS); scales across datasets for industrial inspection and monitoring.

Selected Research Publications

Rossetti, Simone et al. (2022). "Max pooling with vision transformers reconciles class and shape in weakly supervised semantic segmentation". In: *European conference on computer vision*. Springer, pp. 446–463.

Sama, Nico et al. (2023). "A new large dataset and a transfer learning methodology for plant phenotyping in Vertical Farms". In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 540–551.

Rossetti, Simone and Fiora Pirri (2024). "Hierarchy-agnostic unsupervised segmentation: parsing semantic image structure". In: *Advances in Neural Information Processing Systems* 37, pp. 98898–98935.

Rossetti, Simone et al. (2026). "CABBAGE: Comprehensive Agricultural Benchmark Backed by AI-Guided Evaluation". In: *Ongoing Work*. URL: https://huggingface.co/datasets/deepplants/cabbage.

Selected Technical Skills

Multimodal & Vision-Language	Vision-Language Models (CLIP, BLIP, Qwen-VL/LLaMA/GPT), Vision-Language-Action (PaLM-E, RT-X, GR00T-N1), Segment Anything (SAM), Object Detection, Instance Segmentation, Action Recognition, Multimodal Fusion
Learning Paradigms	Weakly- and Unsupervised Segmentation, Self-Supervised (DINO, SwAV, SeLA), Contrastive (SimCLR, MoCo, BYOL), Masked Modeling (BERT, MAE), Diffusion, VAEs, GANs
Training & Distributed Optimization	Multi-GPU (PyTorch DDP, FSDP, DeepSpeed ZeRO, Megatron-LM TP), LoRA, PyTorch Lightning, Hugging Face, xFormers
Evaluation & Benchmarking	Benchmark Design, Ablation Studies, Cross-Dataset Robustness, Weights & Biases, MLflow, Hydra, OmegaConf
Model Serving & Agentic Integration	LangGraph, LangChain, LlamaIndex, SGLang, Ollama, vLLM, Milvus, Qdrant, Multimodal RAG, Docker, FastAPI, CI/CD
Tooling & Workflow Automation	Python, PyTorch, TIMM, OpenCLIP, Torchvision, OpenCV, Albumentations, TensorBoard, Git, Linux

Additional Information

Languages	Italian (Native), English (Fluent)
Poster Sessions	NeurIPS '24, ICCV '23, ECCV '22
Networking & Partnerships	EU Research & Innovation Days '25
Residential Trainings	ICVSS '22, DeepLearn '22